

Introduction to Proof

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Proof

A proof is a method for establishing the truth of a statement.

- A mathematical proof is a verification for establishing the truth of a **proposition** by a chain of **logical deductions** from a set of **axioms**

Proposition	Axioms	Logical Deductions
• A proposition is a statement, or declarative sentence that can either be true or false but not both	• Statements which are given true • Example: • For mathematical quantities a and b, if $a = b$, then $b = a$	• Logical deduction is a process of reasoning in which conclusions are drawn based on given premises

Types of Proof

1. Direct Proof
2. Disproof by Counter Example
3. Proof by Contradiction
4. Proof by Contrapositive

Direct Proof

- A direct proof is defined as

Assume that p is true. Use rules of inference, axioms, and logical equivalences to show that q is also true.

- In other words

A conditional statement $p \rightarrow q$ is constructed

1. when the first step is the assumption is that **p is true**
2. Subsequent steps may be constructed by **rule of inference**
3. the final step show that **q must also be true**.

Example of Direct Proof

- Given a direct proof of theorem

if n is odd integer then n^2 is also odd

Solution:

$$\forall n (p(n) \rightarrow Q(n))$$

$$P(n) = n \text{ is odd integer}$$

$$n = 2k + 1$$

Now we want to show n^2 is also odd integer

$$n = 2k + 1$$

Taking the square root on both sides

$$\underline{n^2} = \underline{(2k + 1)^2}$$

$$\underline{n^2} = \underline{4k^2 + 4k + 1}$$

Example (Cont...)

- Now we can say that

$$P(n) = 2k + 1$$

$$Q(n) = 4k^2 + 4k + 1$$

Suppose that $k = 6$, that is an even integer

$$P(6) = 2 \cdot 6 + 1 = 13$$

$$Q(6) = 4 \cdot (6)^2 + 4 \cdot (6) + 1 = 169$$

Example 2:

- Given a direct proof

if m and n are both perfect squares, then nm is also perfect square.

Solution:

$$\forall n (p(n) \rightarrow Q(n))$$

$P(n)$ = m and n are perfect square

$Q(n)$ = nm and also perfect square

Let suppose

$$m = \underline{s}^2 \text{ and } n = \underline{t}^2$$

The goal is to proof that nm is also perfect square, by the definition of commutative law of multiplication

$$\mathbf{mn} = (\mathbf{st})^2$$

Example 3:

For all integers a and b , if a and b are positive and a divides b , then $a \leq b$.

Solution:

Then there exists an integer k such that,

$$B = a * k$$

Since both a and b are positive, it follows
that

$$1 \leq k$$

Multiplying both sides by a we get

$$a \leq a * k = b$$

$$a \leq b$$

Proof by Contrapositive

- A statement and its contrapositive are logically equivalent. This means that *if we prove $\sim Q \Rightarrow \sim P$, we've also proven $P \Rightarrow Q$* . This is called contrapositive proof.
- It proves $P \Rightarrow Q$ by a direct proof of the contrapositive statement $\sim Q \Rightarrow \sim P$.

Example 1:

If r is irrational, then $\sqrt{r}$ is also irrational

Solution:

Let suppose

$P(x) = r$ is rational

$Q(x) = \sqrt{r}$ is also irrational

The Conditional statement will be

$$(p(n) \rightarrow Q(n))$$

Its Contrapositive will be

$$\sim Q(n) \rightarrow \sim P(n)$$

Example (Cont..)

- If n is irrational then its opposite will be rational therefore

$$\sqrt{r} = \frac{m}{n}$$

Taking the square root both sides

$$r = \frac{m^2}{n^2}$$

Since m^2 and n^2 are integers, so r is rational

Proof by Contradiction

- Another indirect form of proof
 - To prove p , assume $\neg p$
 - Derive a contradiction q such that $\neg p \rightarrow q$
 - This proves that p is true
 - Why is this reasoning valid:
 - A contradiction (e.g., $r \wedge \neg r$) is always F , hence $\neg p \rightarrow F$
 - Since $\neg p \rightarrow F$ is true, the contrapositive $T \rightarrow p$ is also true

Example

Proof by contradiction that $\sqrt{2}$ is irrational

Proof: We use proof by contradiction. Suppose 2 is rational.

Then there exist integers m and n with no common factors such

$$\sqrt{2} = \frac{m}{n}$$

Taking the square root both sides

$$2 = \frac{m^2}{n^2}$$
$$2 n^2 = m^2$$

This means that m^2 is even, which implies that m is even. That is for some integer k.

Example (Cont...)

$$m = 2k$$

$$m^2 = (2k)^2 = 4k^2 = 2n^2$$

Or equivalently

$$n^2 = 2k^2$$

Example 2:

Prove that if $3n+2$ is odd, then n is odd (n is an integer).

Solution:

$p \equiv 3n+2 \text{ is odd}$

$q \equiv n \text{ is odd}$

Assume: **$p \wedge \neg q$**

$3n+2 \text{ is odd} \wedge n \text{ is even}$

Since n is even, $n = 2k$ for some integer k . Thus

$$3n+2 = 3(2k)+2 = 6k+2 = 2(3k+1) = 2j \text{ for } j = 3k+1$$

Therefore $3n+2$ is even

This contradicts the assumption that $3n+2$ is odd

Since the assumption implies F, the original statement is T:

If $3n+2$ is odd then n is odd

Disproof by Counter Example

- Disproving a statement by a single example is called disproof by counter-example.
- when we disprove our statement is called a counterexample.

Example:

Disproof by counterexample, that $x^2 - x + 5 = 0$ is prime for every x , where x belongs to a set of integers.

Example:

According to the statement, $x^2 - x + 5$ gives the prime values whenever we put any value of x .

Example (Cont...)

- Here we have to check on different values, getting much information is best to state results.

Let us start from $x = -2$, and put the value of x in above equation

$$(-2)^2 - (-2) + 5$$

$$4+2+5$$

$$11$$

- Since 11 is prime, this is not a counterexample.

Example (Cont...)

- Here we have to check on different values, getting much information is best to state results.

Let us start from $x = 3$, and put the value of x in above equation

$$(3)^2 - (3) + 5$$

$$9 - 3 + 5$$

$$11$$

- Since 11 is prime, this is not a counterexample.

THANKS